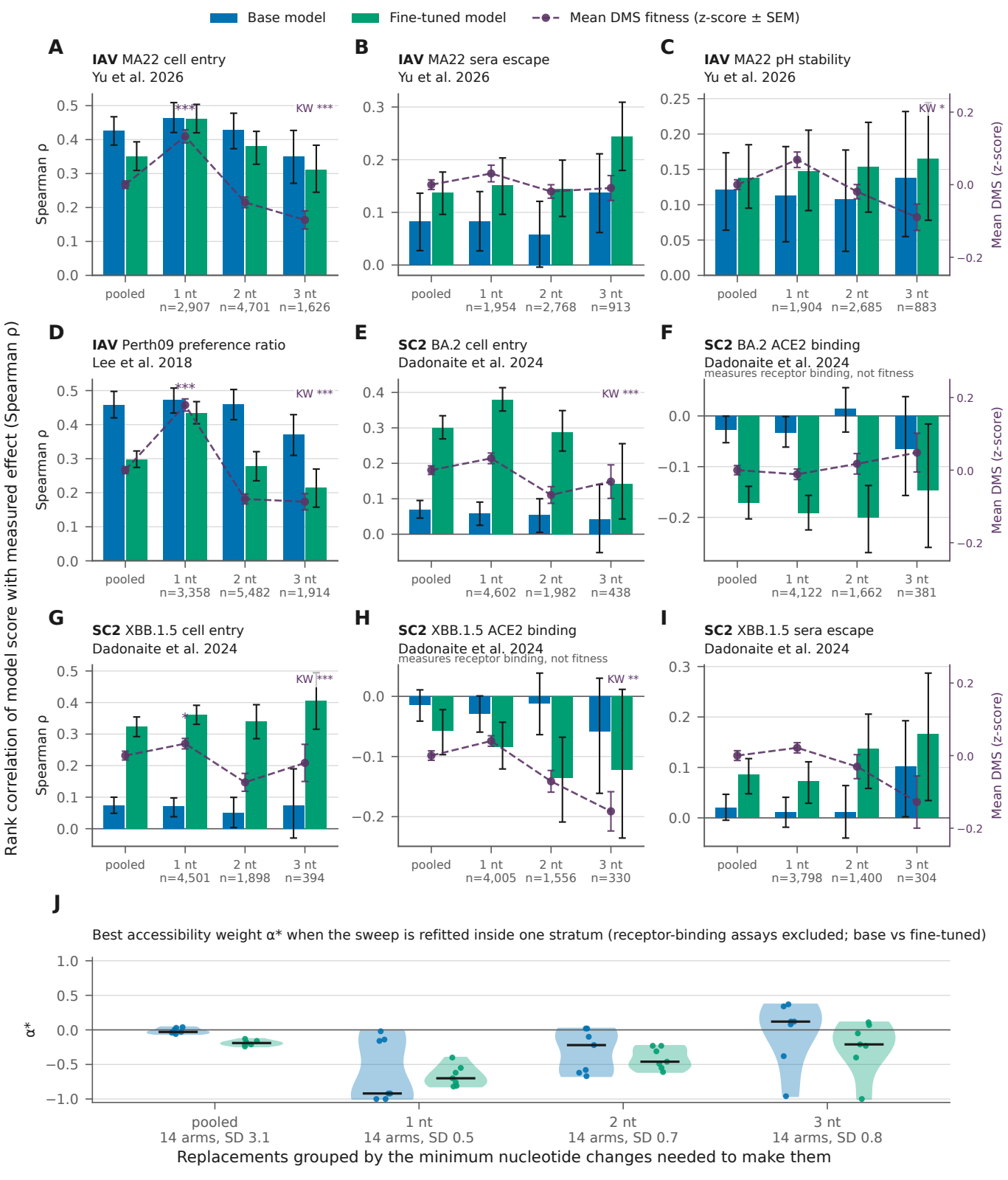
Base model
Fine-tuned model
Mean DMS fitness (z-score ± SEM)
Rank correlation of model score with measured effect (Spearman ρ)
Mean DMS (z-score)
A
IAV MA22 cell entry
Yu et al. 2026
KW ***
B
IAV MA22 sera escape
Yu et al. 2026
C
IAV MA22 pH stability
Yu et al. 2026
KW *
D
IAV Perth09 preference ratio
Lee et al. 2018
KW ***
E
SC2 BA.2 cell entry
Dadonaite et al. 2024
KW ***
F
SC2 BA.2 ACE2 binding
Dadonaite et al. 2024
measures receptor binding, not fitness
G
SC2 XBB.1.5 cell entry
Dadonaite et al. 2024
KW ***
H
SC2 XBB.1.5 ACE2 binding
Dadonaite et al. 2024
measures receptor binding, not fitness
KW **
I
SC2 XBB.1.5 sera escape
Dadonaite et al. 2024
Spearman ρ
pooled
1 nt n=2,907
2 nt n=4,701
3 nt n=1,626
1 nt n=1,954
2 nt n=2,768
3 nt n=913
1 nt n=1,904
2 nt n=2,685
3 nt n=883
1 nt n=3,358
2 nt n=5,482
3 nt n=1,914
1 nt n=4,602
2 nt n=1,982
3 nt n=438
1 nt n=4,122
2 nt n=1,662
3 nt n=381
1 nt n=4,501
2 nt n=1,898
3 nt n=394
1 nt n=4,005
2 nt n=1,556
3 nt n=330
1 nt n=3,798
2 nt n=1,400
3 nt n=304
J
Best accessibility weight α* when the sweep is refitted inside one stratum (receptor-binding assays excluded; base vs fine-tuned)
α*
pooled
14 arms, SD 3.1
1 nt
14 arms, SD 0.5
2 nt
14 arms, SD 0.7
3 nt
14 arms, SD 0.8
Replacements grouped by the minimum nucleotide changes needed to make them